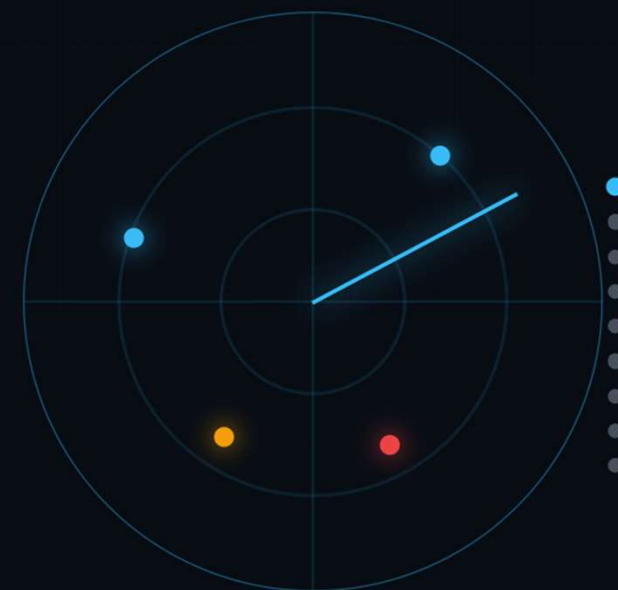


IBM BOB AI INNOVATION HACKATHON 2026

TEAM BYTEFORGE · AI TRACK

# THREAT INTELLIGENCE CORRELATION & ALERT PRIORITISATION ASSISTANT

Turn fragmented signals into one commander-ready threat picture.



## THE PROBLEM

# ANALYSTS DROWN IN ALERTS. REAL THREATS HIDE IN THE NOISE.

02

### 01 · SIEM

#### THOUSANDS OF ALERTS

Events arrive with different formats, priorities and context.



### 02 · NOISE POOL

#### SIGNAL GETS BURIED

One critical pattern competes with false positives.



### 03 · ANALYST

#### ATTENTION IS LIMITED

Manual triage slows investigation and response.

#### REAL THREAT MISSED

Catastrophic impact

#### FALSE POSITIVES CHASED

Wasted analyst hours

Thousands of signals. Limited analyst attention.

## THE SOLUTION

# ONE PIPELINE. FOUR STAGES. ONE CLEAR ANSWER.

03



**CORRELATE → PRIORITISE → UNDERSTAND → PREDICT → BRIEF**

## CORE CAPABILITIES

# FROM RAW ALERTS TO ACTIONABLE INTELLIGENCE.

### 01 • MULTI-SOURCE CORRELATION

Turns fragmented alerts into incidents.  
SIEM • Sensors • Intel • Satellite

### 02 • DYNAMIC RISK SCORING

Ranks what needs attention first.

0 — 100

### 03 • THREAT DNA

Behavioural fingerprint beyond IPs & malware.

### 04 • MITRE ATT&CK

Maps observed behaviour to attacker techniques.

Less alert noise. More meaningful incidents.

# DON'T JUST DETECT THE ATTACK. UNDERSTAND WHAT COMES NEXT.

05

CYBER SIGNALS

SATELLITE / GEOSPATIAL SIGNAL

AI-ASSISTED REASONING

ACTIVE  
THREAT

## CYBER-PHYSICAL THREAT FUSION

Observe → Correlate → Understand

### LIKELY NEXT TECHNIQUE

Predictive kill-chain forecasting with uncertainty — not guaranteed outcomes.



### AI INVESTIGATION ASSISTANT

Ask → Investigate → Explain

### BLUF REPORTING

Threat · Impact · Confidence · Evidence · Next steps

COMMAND CENTER

- ☐ Overview
- ☒ Incidents
- ☐ Detection & Scoring
- ☐ Context Intelligence
- ☐ AI Reasoning

FILTERS

Priority

Critical ×

High ×

Medium ×

Low ×

× ⌵

Status

new ×

investigating ×

closed ×

× ⌵

Source

intel ×

satellite ×

sensor ×

siem ×

× ⌵

• SYSTEM OPERATIONAL

THREAT MONITORING ACTIVE

DATA SOURCES

|           |          |
|-----------|----------|
| INTEL     | 2 alerts |
| SATELLITE | 2 alerts |
| SENSOR    | 2 alerts |

DETECTION & SCORING

CONTEXT INTELLIGENCE

AI REASONING

DETECTION & SCORING

RISK ASSESSMENT

RISK SCORE

97.5

/ 100



PRIORITY

Critical

STATUS

NEW

> Why this score?

FALSE POSITIVE ASSESSMENT

✓ NO — Genuine Threat

Incident flagged as genuine threat: high alert volume (9 alerts); 7 high/critical severity alert(s); 2 suspicious event type(s); repeated suspicious events: login\_failure; alerts from multiple sources: intel, satellite, sensor, siem; high-value asset(s) targeted: server-01; 4 alert(s) with suspicious payload content

THREAT DNA SIGNATURE

SEV\_CRITICAL|MULTI\_SOURCE\_3PLUS|MALWARE|SCRIPT\_EXECUTION#04aff8773

HISTORICAL CONTEXT — SIMILAR INCIDENTS

No similar historical incidents.

ALERT BREAKDOWN

CRITICAL

4

HIGH

3

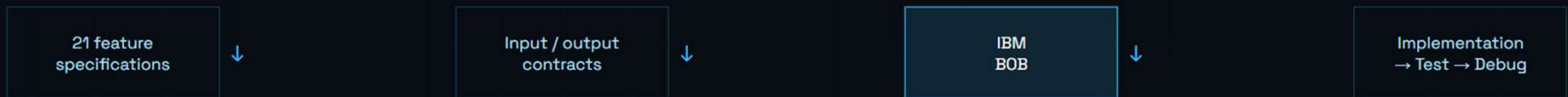
LOW

2

> View all 9 alert(s)

# IBM BOB HELPED TURN OUR SPECIFICATIONS INTO A WORKING SYSTEM.

07



↻ Refine with human review ↻

## 01 • WE DEFINED

21 features · 4 modules · shared contracts

## 02 • BOB ACCELERATED

Python implementation · debugging · UI refinement

## 03 • HUMANS VALIDATED

Testing · review · correction · integration

AI-ASSISTED DEVELOPMENT. HUMAN-LED ENGINEERING.

# FROM PROTOTYPE TO OPERATIONAL INTELLIGENCE.

08

## STAGE 1 · NOW

### WORKING PROTOTYPE

- Multi-source ingestion
- Alert correlation
- Risk prioritisation
- MITRE context
- Cyber-physical fusion

## STAGE 2 · NEXT

### OPERATIONAL INTEGRATION

- Real SIEM APIs
- Sensor feeds
- Persistent Threat DNA
- Real-time intelligence feeds
- Larger-scale correlation

## STAGE 3 · FUTURE

### DEFENCE-SCALE INTELLIGENCE

- Continuous monitoring
- Predictive threat intelligence
- Faster analyst response
- Commander decision support

LESS NOISE. FASTER ANALYSIS. BETTER DECISIONS.

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THREAT INTELLIGENCE CORRELATION & ALERT PRIORITISATION ASSISTANT

# THANK YOU

GitHub: <https://github.com/25cs065-ux/bob-ai-hackathon-ByteForge>

Demo: <https://youtu.be/RB4gdMBzCgk>

PATEL KHUSHI – 25CS065 – FRONTEND AND INGESTION MODULE

PATEL NANDANI – 25CS073 – CONTEXT MODULE

PANCHAL SIDDHI -- 25CE066 – DETECTION MODULE

DEKIVADIYA YATRI -- 25CE021 – REASONING MODULE